Town of Ladysmith

STAFF REPORT

To: Ruth Malli, City Manager
From: John Manson, Peng, FEC
Date: April 2, 2014
File No:

Re: 2013 WATER REPORT AND FILTRATION DECISION

RECOMMENDATION(S):

That Council:

1. Receive the draft 2013 Water Report for information;

2. Direct staff to pursue, in principle, water filtration as the method of secondary water treatment for the Town of Ladysmith, and waive the purchasing policy for a Water Filtration Pilot Study; and,

3. Direct staff to amend capital plans to provide for a secondary water filtration system for the Town.

If Council wishes, they may invite representatives from Island Health (VIHA) to a future meeting to discuss and answer any questions about water regulatory requirements.

PURPOSE:

The purpose of this report is to update Council on the current status of the Town's Water System, and provide options for moving forward to meet our Water Permit commitments under the Water Act.

INTRODUCTION/BACKGROUND:

The Town's water system is operated and maintained in accordance with a permit issued through Island Health (VIHA).

As part of our permit requirements, the Town is required to prepare and publish water quality reports. The Town's 2013 Water Report will be presented at the Council meeting and afterwards posted on the Town's website. The report confirms

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that the Town's water system generally complies with the intent of our permit, aside from the question of water treatment, the subject of this report.

Since the late 2000's, Ladysmith, as with most other municipalities in BC, have been directed by the Province to upgrade domestic surface water systems to meet the following water treatment requirements:

- 4 log removal/inactivation of viruses;
- 3 log removal/inactivation of Giardia cycts and Crypotsporidium oocysts;
- 2 treatment processes, usually filtration and disinfection;
- 1 NTU turbidity (maximum) in finished water

The Health Authority (VIHA) also requires that all sources of raw water prior to treatment be relatively free of E-Coli bacteria (less than 10% of samples in any 6 month period can be over 20 cfu/100mL), and upstream watersheds must be managed in a manner consistent with the goal of maintaining a 'healthy' water source.

If all of the above noted parameters are in place, VIHA may issue a 'filtration deferral', which provides for the operation of a minimum of two water treatment processes, typically UV disinfection, ozone disinfection, and chlorination.

The Town has been actively upgrading the treatment and management of our water supplies in order to meet the 4-3-2-1 policy noted above. The Town has previously completed a major conversion of an old open reservoir to a concrete covered reservoir that is sufficient to meet the Towns needs for the immediate future. The Town also recently completed a dedicated water main 'connection' of the Stocking Lake raw water supply to the new reservoir, which will allow all surface water sources to be treated and stored in one location. Most recently, the Town commenced the upgrading of the old chlorination facility, now located at the old open reservoir site.

The Town has also been closely monitoring water quality parameters, such as source water turbidity, to determine if the Town could continue to pursue a filtration deferral. In particular, turbidity samples have been compiled for Stocking Lake, Holland Creek at Chicken Ladder, and, most recently, Holland Lake. The Town now has over 1 year of continuous turbidity data for Holland Lake, including the winter months when the Lake is frozen over.

Stocking Lake consistently delivers high quality water, significantly less than 1 NTU, the 'maximum' allowable turbidity allowance for filtration deferral, year round. In 2013, Holland Creek (Chicken Ladder) was over 1 NTU for 13 days (less than the maximum 18 days permitted for filtration deferral), which is considerably better than previous history at Chicken Ladder.

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The poor turbidity values in Holland Creek are primarily the result of sudden rainfall events; it is also impacted by natural and human activities in the watershed. The result is rapid deterioration of water quality at this location during periods of heavy rainfall. Compounding this is the relative lack of water storage capacity at the Holland Creek intake, which does not allow any blending of poorer quality water with higher quality, which typically occurs in larger lakes and reservoirs.

Telemetry data is collected at the Public Works office, and for the last decade or more the Town rapidly switches water sources as the situation warrants. Typically the Town will draw its water from Holland Creek, except in the winter months when Stocking Lake is overflowing. The Town will switch back to Stocking during high turbidity events, returning back to Holland Creek after the event has passed and the water returned to under 1 NTU. To date the Town has been able to successfully keep our 'blended' water below the 1 NTU level for essentially all of the year in 2013.

Part of the Town's strategy moving forward was to provide a direct connection from Stocking Lake to Holland Lake, such that expected 'higher' quality Holland Lake water could also accessed when Holland Creek was not suitable for use. However, 2013 turbidity readings taken from Holland Lake have provided some surprising results: Holland Lake water exceeded the 1 NTU level for about 90 days in 2013, mostly during the winter months when the lake was frozen over. The turbidity 'event' did not subside until spring, when the ice came off the lake.

The Holland Lake results have provided us with an opportunity to reconsider the present approach to filtration deferral, which was:

- Completion of the Chlorination system upgrade;
- Construction of a secondary treatment process, such as UV disinfection or ozone disinfection;
- Management of water sources to enable higher quality Stocking Lake water to be used subject to water licensing restrictions;
- Construction of a new supply main to Holland Lake to provide backup 'high' quality water during turbidity events at Holland Creek.
- Continuous monitoring of turbidity levels in all sources of supply, whether they are in use or not;
- Increased (or more effective) management of the watershed;
- Ongoing water quality testing of the raw water supplies.

The above noted approach recognises the significant volume contribution that the Holland Creek supply provides to the Town – the Town gets more than half of its water supply needs from Holland Creek, and we expect that this proportion will increase as water use gradually increases over the coming decades.

The Town is also in the process of completing an update to previous watershed modeling work that was undertaken in 2008. This work will include the effects of

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climate change, and will provide strategic direction on long term water supply for the Town. We also expect that this study will provide updated information that will enable the Town to clarify existing water licenses, particularly on Stocking Lake. This may entail making application to the Province for amendments to the Stocking License, including storage.

While it would be advantageous to complete the watershed modelling and licensing work before making a decision on major improvement works such as the Holland to Stocking pipeline, we wish to seek Council endorsement to consider proceeding with filtration. The rational for filtration is as follows:

- The poor turbidity results on Holland Lake bring into question the assumption that Holland Lake can be used as a reliable low turbidity source at all times of the year, and this option is not feasible until the Holland to Stocking pipeline is constructed;
- If the Holland to Stocking pipeline is postponed, the cost of filtration is about the same capital cost of alternative secondary disinfection (UV/Ozone) plus the Holland to Stocking Pipeline;
- Should the Town proceed with UV/Ozone disinfection at this time, and filtration is required in the future, either as a result of changing regulation or due to a change in the Towns ability to meet the existing standard (for example, weather related or activities in the watershed that impact water quality, some of which are beyond the Towns' control) then the cost of the UV/Ozone system will not be a good investment;
- Filtration deferral is not recommended for the long-term. Control of the watershed is a critical component of filtration referral. Even with control of the watershed, other communities (such as Vancouver) still choose filtration partially because natural events cannot be controlled, even if human activity can be restricted;

Should Council agree with the recommendation to proceed with filtration, a pilot project will be required to provide actual test data on our water sources, this will form the basis of a full scale design. We have tentatively placed this project in the Capital Plan for 2014, at a cost of $165,000. Following this pilot project, we could initiate design in 2015, and commence construction following the design process, with completion in late 2016 or early 2017. The timing of these projects will largely be driven by grant availability. These projects have also tentatively been placed in the Capital Plan for Council consideration in the current financial planning process. The Holland to Stocking Pipeline also remains in the Capital Plan, for review later this year, following the completion of the Watershed Study. If the Watershed Study indicates that the Holland to Stocking Pipeline is essential regardless of filtration then it will remain in the Capital Plan. If the Watershed Study indicates that with filtration, the Holland to Stocking Pipeline is not required, then it can be removed.

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It should also be noted that the design and construction components of the project will likely require borrowing, and hence the assent of the electors will be required.

SCOPE OF WORK:

The scope of work is the construction of a secondary disinfection system for the Towns water supply. The recommendation in this report is the consideration of a change to the current direction of filtration deferral. Currently the direction is to construct a UV or Ozone system in lieu of filtration.

The scope of work will involve several phases, starting with the consideration of the change of direction to modeling and pilot study through to final construction.

ALTERNATIVES:

The Town could continue to pursue filtration deferral. This would require further discussions with Ministry of Health officials to determine if the current turbidity and microbiological raw water sampling support this approach.

FINANCIAL IMPLICATIONS:

The cost of the Filtration Pilot Project is included in the 2014 Capital Plan.

Borrowing will be required to fund the construction of the Filtration project. An increase in the parcel tax will be necessary to service the debt required for this project.

LEGAL IMPLICATIONS:

n/a

CITIZEN/PUBLIC RELATIONS IMPLICATIONS:

The public will likely be involved in providing input into the borrowing process to support for loan authorization bylaw. Although the question of filtration is largely driven through the regulatory process, it is recommended that the Town actively engage citizens in this process.

INTERDEPARTMENTAL INVOLVEMENT/IMPLICATIONS:

The project will principally be managed by public works staff. However, the project will impact finance from a budgeting and resource management point of view.

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RESOURCE IMPLICATIONS:

The significant size of the project will require the use of outside consultants and contractors. Due to time limitations, we wish to retain Koers and Associates to manage the Filtration Pilot project; consultant selection for other components will follow later in the year.

ALIGNMENT WITH SUSTAINABILITY VISIONING REPORT:

Provision and protection of water aligns with the Sustainability Vision.

ALIGNMENT WITH STRATEGIC PRIORITIES:

Aligns with the enhanced infrastructure priority.

SUMMARY:

The Town's water system is regulated through the Water Act, and administratively through Island Health (VIHA). A few years ago, VIHA published new guidelines for the treatment of surface water supplies for domestic consumption. The new guidelines require filtration of surface water as one of the treatment steps, in order to meet biological/pathogenic organism reductions outlined in the guidelines. However, the guidelines provide for an exemption of filtration, provided that turbidity levels stay below 1 NTU all but 18 days in any year, among other criteria. To determine the ability to meet these exemption criteria, the Town has been closely monitoring results for the past year.

The turbidity testing was done for the full year of 2013 and the results indicate that the Town will likely have difficulty meeting VIHA criteria for filtration avoidance. Staff are therefore recommending that Council consider proceeding with a Filtration Pilot Project in 2014; and plan for the completion of filtration construction in the current Financial Plan.

I concur with the recommendation.



Ruth Malli, City Manager

ATTACHMENTS:
2013 Annual Water Report (to be distributed at the Council presentation)

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